

Karthik D K

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Summary

Firmware Engineer building the RISC-V software ecosystem for automotive and IoT at Bosch's TrusteD-V Innovation Team. 6+ years of embedded systems experience spanning kernel development, RTOS porting, and automotive ECU application software. Expertise in Rust, C, and Python for bare-metal and safety-critical systems. RISC-V Summit 2025 featured contributor.

Experience

Firmware Engineer *Bosch Global Software Technologies* *August 2019 - Present*

Hubris OS Port to RISC-V 🔄

- Ported Hubris OS to RISC-V - kernel primitives, trap/exception handlers, and PMP-based memory isolation for the Hazard3 core on Raspberry Pi Pico 2.
- Enabled dual-architecture support across ARM **Cortex-M33** and RISC-V **Hazard3** via custom linker scripts and startup code; worked around RP2350 errata and validated bring-up using OpenOCD/GDB over the Raspberry Pi Debug Probe.

TrusteD-V RTOS

- Developed proof-of-concept RTOS port for Mindgrove Secure IoT development board with working QEMU emulation support.
- Debugged kernel bring-up issues using Segger J-Link with OpenOCD and GDB.

Rust SDK for CDAC VEGA Processor

- Created the first Rust bare-metal ecosystem for the CDAC VEGA ARIES v3 board (THEJAS32 RISC-V SoC).
- Reverse-engineered register maps from the VEGA C SDK into a Python-based SVD generator, producing type-safe PACs for 15+ peripherals (UART, SPI, I2C, Timer, GPIO, ADC, PWM, PLIC) validated with bare-metal examples.

SiFive E34 Board Bring-up (Trina-Pi)

- Performed complete bare-metal bring-up of the UpBeatTech Trina-Pi board (SiFive E34 RISC-V core) entirely in Rust.
- Implemented a UART peripheral driver from scratch for serial console and debug I/O.

MLLib DecisionTree Inference

- Built a lightweight C DecisionTree inference engine for a resource-constrained PowerTrain ECU, using FlatBuffers for model serialization and calibration-tool parameterization.

Advanced Exhaust Temperature Management (Onsite — Germany, 3 months)

- Developed a CAN-based temperature-sensor component with diagnostics and statistical catalyst monitoring to meet EU7 emission standards.

Skills

Programming: Rust, C, Python

RTOS/Kernel: Hubris OS, xv6, bare-metal programming

Debug Tools: OpenOCD, GDB, tio, RP Debug Probe, Segger J-Link

Development Tools and Frameworks: WSL, Git, Claude Code, PyTorch, Cargo, Make, Just, ASCET

Certifications

Machine Learning: Stanford University (Andrew Ng)
Coursera

Certificate
2020

Achievements

- **RISC-V Summit 2025:** Trusted-V project contributor (featured talk). 
- **Bosch Recognition:** Best Developer Award, Crowdsourcing Champ, AI Hackathon Winner. 
- **Chess:** State-Level Championship participant.

Education

University Vishweshwaraya College of Engineering, Bengaluru 75.2%
B.E in Electronics and Communication 2019

Devaraja URS PU College, Doddaballapur 94.5%
PUC 2015

Blogging

- Writing on firmware, systems programming, Rust, and motorcycles - a public knowledge repository that documents my work and helps others (and my future self). 

Open Source Projects

Contributions

- Active open-source contributor with merged pull requests into oreboot (Rust boot firmware) and the fastai and FluxML deep-learning libraries.

robot-hat-rs

- Developed no_std Rust drivers (I2C, PWM, ADC) and a hardware abstraction layer for motor control, servo actuation, and analog sensor reading on the Robot HAT platform.
- Published on crates.io, receiving 4,000+ downloads from the community.

ixv

- Developed a Rust CLI for verifying Intel HEX files. Published on crates.io with 4,000+ downloads.

picars

- Created an autonomous vehicle system using Raspberry Pi and PiCar-X kit, leveraging Rust and Python.
- Developed Rust bindings and interfaced with Python to optimize execution speed.

Languages

- English, Kannada, Telugu